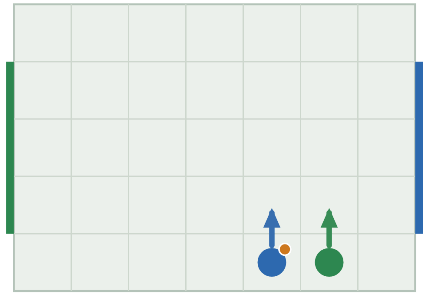
SOCCER MARKOV GAME · RESEARCH MEETING FOLLOW-UP

Player positions and the 4×4 Q matrix

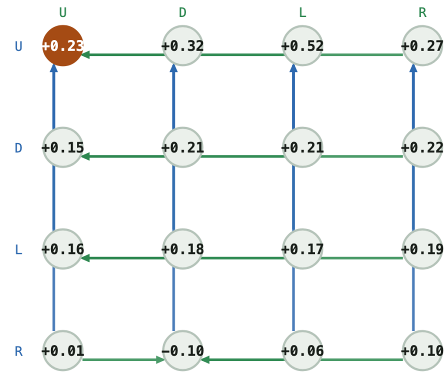
At the research meeting the professor drew a stage game by hand: a small grid of cells, each marked, with arrows chasing each other around the grid to show that no cell is a stable outcome, at players positioned `(0, 0)` and `(1, 1)`. This is that sketch, generated from the exact solver instead of drawn free-hand, paired with a picture of where the two players actually are, and the exact, complete 4×4 `{U, D, L, R}` **Q matrix** underneath every case — always the full grid, never reduced, per the professor's own correction that the matrix should stay 4×4 and that the matrix itself, not an entropy number, is the actual output.

A pure state -- one safe cell, nothing to guess



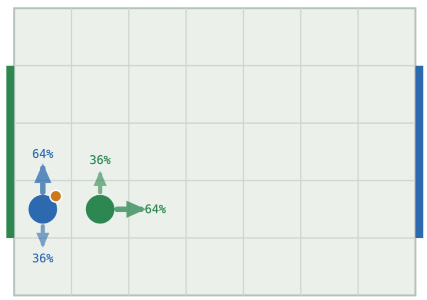
pure equilibrium · V = +0.234

Q matrix -- (4, 0, 5, 0, 0)



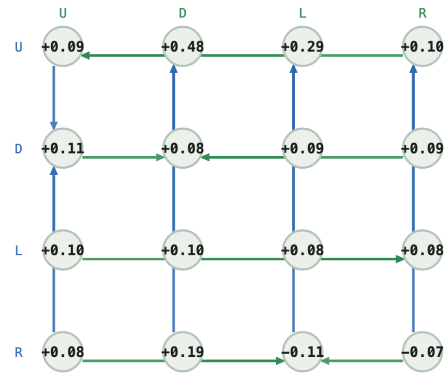
pure saddle at U/U

The typical mix -- which goal row to head for



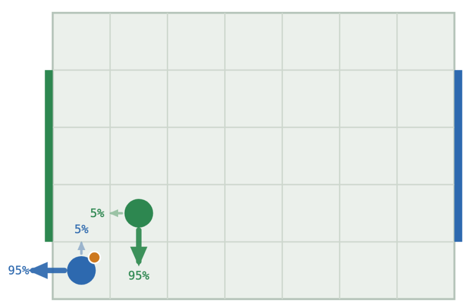
mixed equilibrium · V = +0.094

Q matrix -- (0, 1, 1, 1, 0)



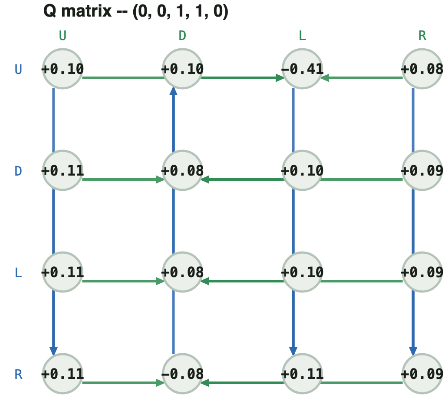
arrows cycle -- no cell is a stable outcome, must mix

The professor's own position -- (0,0) and (1,1)



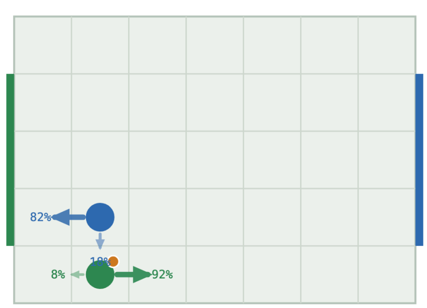
mixed equilibrium · V = +0.078

Q matrix -- (0, 0, 1, 1, 0)



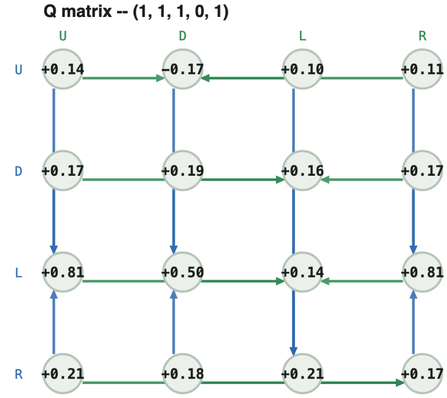
arrows cycle -- no cell is a stable outcome, must mix

A clean L/R indifference example



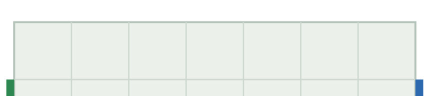
mixed equilibrium · V = -0.206

Q matrix -- (1, 1, 1, 0, 1)

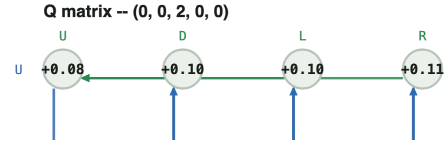


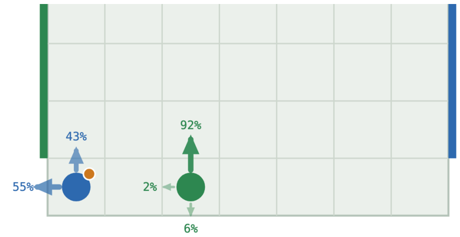
arrows cycle -- no cell is a stable outcome, must mix

A genuine 3-action mix, not a 2-cycle



Q matrix -- (0, 0, 2, 0, 0)



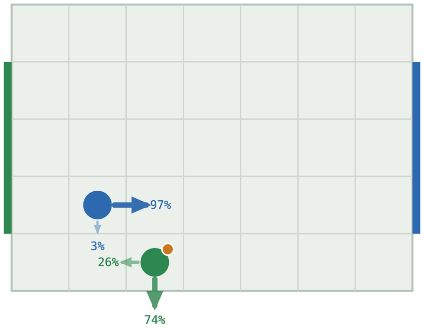


mixed equilibrium · V = +0.085

D	+0.09	+0.08	+0.08	+0.09
L	+0.09	+0.08	+0.08	+0.09
R	+0.09	+0.10	-0.09	+0.11

arrows cycle -- no cell is a stable outcome, must mix

A near-pure hedge -- where rounding would lie



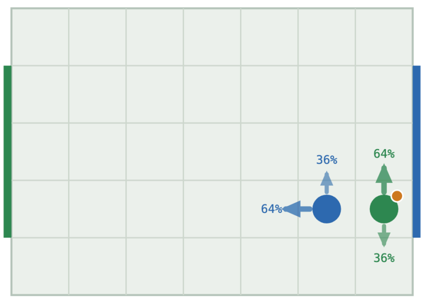
mixed equilibrium · V = -0.168

Q matrix -- (1, 1, 2, 0, 1)

	U	D	L	R
U	+0.32	+0.32	+0.25	-0.18
D	+0.21	+0.21	+0.21	+0.17
L	+0.17	+0.04	+0.16	+0.17
R	+0.18	+0.18	+0.18	+0.14

arrows cycle -- no cell is a stable outcome, must mix

The typical mix, mirrored -- same shape, value negated



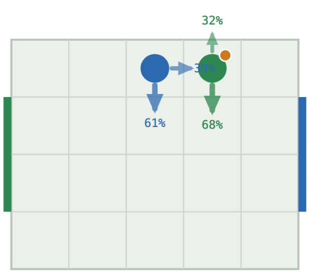
mixed equilibrium · V = -0.094

Q matrix -- (5, 1, 6, 1, 1)

	U	D	L	R
U	+0.09	+0.48	+0.10	+0.29
D	+0.11	+0.08	+0.09	+0.09
L	+0.08	+0.19	-0.07	-0.11
R	+0.10	+0.10	+0.08	+0.08

arrows cycle -- no cell is a stable outcome, must mix

This project's own tackle rule -- a different mechanism, same duel



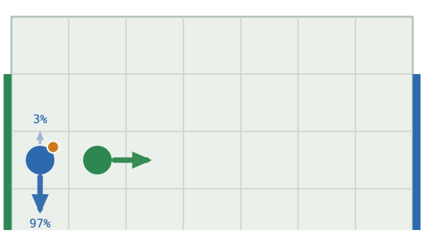
mixed equilibrium · V = +0.013

Q matrix -- (2, 3, 3, 3, 1)

	U	D	L	R
U	-0.01	+0.01	+0.02	-0.04
D	+0.07	-0.02	+0.01	+0.00
L	-0.02	-0.08	+0.02	-0.04
R	-0.01	-0.03	-0.01	-0.04

arrows cycle -- no cell is a stable outcome, must mix

The asymmetric mix -- the carrier still guesses, the defender doesn't



Q matrix -- (0, 2, 1, 2, 0)

	U	D	L	R
U	+0.08	+0.48	+0.29	+0.10
D	+0.48	+0.08	+0.29	+0.10
L	+0.09	+0.09	+0.09	+0.09

All nine cases: board position (left) and the full 4×4 Q matrix redrawn as a node-and-arrow graph (right). A lit node with no outgoing arrow is a pure saddle; a closed loop of arrows means no cell is safe.

Reading the graph

Each node is one cell of the complete 4×4 Q matrix (`row = the carrier's action`, `col = the defender's action`, always reoriented so the printed payoff is the carrier's, regardless of which player id actually carries the ball). A blue arrow points from a cell to the cell in the same column with the higher payoff — the carrier's own reason to switch rows. A green arrow points from a cell to the cell in the same row with the *lower* payoff — the defender's reason to switch columns, since the defender minimises. Every case shows all 16 cells, so the shapes are directly comparable.

Case 1 `(4, 0, 5, 0, 0)`

A pure state, for contrast. Saddle at `U/U`. The carrier plays U always; the defender answers U always; neither has any reason to deviate.

	U	D	L	R
U	0.234	0.317	0.523	0.272
D	0.151	0.211	0.211	0.222
L	0.161	0.182	0.172	0.190
R	0.012	-0.095	0.058	0.098

Case 2 `(0, 1, 1, 1, 0)`

The typical mix. Gap 0.0136. The carrier mixes `U 63.5% / D 36.5%`; the defender answers `U 36.5% / R 63.5%`. This is the common shape: the carrier is deciding which goal row to head for, and the defender is guessing which one. **90 of the 94 mixed states on the canonical board cross a vertical pair exactly like this one.**

```
        U       D       L       R
U   0.086   0.478   0.286   0.099
D   0.109   0.076   0.085   0.086
L   0.103   0.103   0.085   0.084
R   0.082   0.187  -0.108  -0.071
```

Case 3 `(0, 0, 1, 1, 0)`

The professor's own position. The board position drawn on the whiteboard: the carrier at `(0, 0)` — the back corner, right against its own goal — and the defender diagonally adjacent at `(1, 1)`. This is the closest match, coordinate for coordinate, to the meeting's own sketch. Distinct from Case 2 (which crosses U/D), this one crosses `U/L` for the carrier and `D/L` for the defender: pinned in the corner, the carrier's two live escapes are straight up the sideline (U) or across along the back line (L), and the defender is guessing which. **Gap 0.0121; the mix is lopsided but genuine** (`U 4.6% / L 95.4%` for the carrier, `D 94.9% / L 5.1%` for the defender) — the corner leaves little room, but not zero.

```
        U       D       L       R
U   0.103   0.103  -0.408   0.084
D   0.109   0.076   0.101   0.086
L   0.109   0.076   0.101   0.086
R   0.112  -0.075   0.112   0.088
```

Case 4 `(1, 1, 1, 0, 1)`

A clean L/R indifference example.

> *"when we move right and when we move left, there's an equal chance of me winning. That's why I am indifferent between the two."*

Gap 0.0445. The carrier's entire live option set is exactly `{L, R}` — no vertical option survives at all, which is what makes this state (rather than Case 3, which crosses U/L) the cleanest match to the meeting's *verbal* description: two actions genuinely equally good against the defender's own mix (`D 18% / L 82%`). **This is the rarer shape overall: only 4 of 94 mixed states cross a purely horizontal pair instead of a vertical one.**

```
       U        D        L        R
U   0.139   -0.173   0.099    0.110
D   0.171    0.185   0.157    0.171
L   0.810    0.500   0.141    0.810
R   0.211    0.181   0.211    0.167
```

Case 5 `(0, 0, 2, 0, 0)`

A genuine 3-action mix. Gap 0.0047, entropy 1.109 bits — the richest mix on this page. Six cells from goal, as far as this board allows, with the defender close enough to threaten three different rows at once: no single row is safely better than the other two. Requested explicitly at the meeting: **the matrix itself, not entropy, is the point.**

```
       U        D        L        R
U   0.084    0.095   0.103    0.110
D   0.086    0.076   0.076    0.086
L   0.086    0.076   0.076    0.086
R   0.088    0.095  -0.088    0.110
```

Case 6 `(1, 1, 2, 0, 1)`

A near-pure hedge, where rounding would lie. Gap 0.0043, entropy only 0.169 bits — the carrier plays `R 97.5%`, `D 2.5%`. Reading 0.975 as 1.0 would be the rounding mistake the meeting flagged: the gap is real and certified, twelve times smaller than a 0.1 rounding grid would resolve.

```
       U       D       L       R
U    0.317   0.317   0.247  -0.178
D    0.211   0.211   0.211   0.167
L    0.171   0.045   0.157   0.171
R    0.182   0.182   0.182   0.135
```

Case 7 `(5, 1, 6, 1, 1)`

The typical mix, mirrored. The board's left-right / role-swap symmetry maps Case 2 exactly onto this state. The values negate to machine precision:

```
V(case 2) = +0.094235
V(mirror) = -0.094235
sum       = -1.39e-17   (zero, to floating-point noise)
```

Direct evidence that Case 2's shape is not a one-off: the identical mix reappears, exactly mirrored, wherever the same relative configuration recurs.

```
       U       D       L       R
U    0.086   0.478   0.099   0.286
D    0.109   0.076   0.086   0.085
L    0.082   0.187  -0.071  -0.108
R    0.103   0.103   0.084   0.085
```

Case 8 `(2, 3, 3, 3, 1), 5×4 board`

This project's own tackle rule. Every other case here gets its coin flip from Littman's random move order. This one is solved under the tackle rule (defender commits to a challenge that wins the ball with fixed probability or bounces off) instead — a mechanism with nothing to do with move order at all. **It produces the same kind of duel** (gap 0.0223): the carrier mixes `D 60.7% / R 39.3%`, the defender mixes `U 32.1% / D 67.9%`. Different cause, same structure.

```
        U        D        L        R
U   -0.012    0.005    0.016   -0.042
D    0.072   -0.022    0.009    0.000
L   -0.019   -0.080    0.019   -0.042
R   -0.007   -0.026   -0.006   -0.039
```

Case 9 `(0, 2, 1, 2, 0)`

The asymmetric mix. Support (2, 1) — gap 0.0095. The odd one out: the carrier's equilibrium still mixes two actions (`U 2.6% / D 97.4%`), but the **defender's** equilibrium is a single fixed move (`R`, 100%). Every other case on this page is a symmetric duel — both players genuinely guessing. Here only one side is.

```
       U        D        L        R
U   0.085    0.478    0.291    0.095
D   0.478    0.085    0.291    0.095
L   0.093    0.093    0.086    0.093
R   0.082    0.082   -0.122   -0.072
```

Why the carrier still needs two actions against a defender who never varies: against the defender's pure `R`, column `R` reads `U 0.095, D 0.095, L 0.093, R -0.072` — **U and D are exactly tied**, both strictly better than L or R. Nothing in the payoffs favours one over the other, so any split of U/D is an equally valid best reply; the solver's LP reports one particular split (2.6% / 97.4%), not a probability forced by the game the way Case 4's 18% / 82% is. This is the meeting's LP-edge-case question made concrete: **a tie between two rows can look identical, in the printed policy, to a genuine forced mix, but it is a different phenomenon** — no best-reply cycle is involved, and either pure U or pure D alone would also be a valid equilibrium reply to the defender's fixed R.

The mathematics behind all nine

A mixed equilibrium is exactly the strategy pair where every action in a player's support earns the **same expected payoff** against the opponent's own mix — that equality is what "indifferent" means in Case 4, and it is also why the best-response graph cycles with no resting point: if one action in the support paid strictly more, that player would raise its weight on it, which is precisely the condition an equilibrium rules out. A best-reply cycle and mutual indifference are the same fact seen from two sides, not two different explanations. Case 9 shows the boundary of that story: indifference alone (a tie against a fixed opponent) can produce the same *symptom* — a fractional policy — without a cycle anywhere in the matrix.

Reproduce with `python scripts/positions.py` (`make positions`) — writes `figures/gallery/positions.svg` and prints every matrix and policy shown here. Full project: `github.com/Charith-Reddy-Pareddy/soccer-markov-nash`.